Nome: William Nunes dos Santos

1

A): $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

B): $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

C): $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

D): $\mathbb{R} \subset \mathbb{C}$

E): $\mathbb{C}$

F): $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

G): $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

H): $\mathbb{R} \subset \mathbb{C}$

2

A) $2+8-3-5+15$

$10-3-5+15$

$7-5+15$

$2+15$

17

B) $12+[35-(10+2)+2]$

$12+[35-12+2]$

$12+[23+2]$

$12+25$

37

/ /

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c) $[(18+3\cdot 2)\div 8+5\cdot 3]\div 6$

$[(18+6)\div 8+5\cdot 3]\div 6$

$[24\div 8+5\cdot 3]\div 6$

$[3+15]\div 6$

$18\div 6$

3

D) $37+[-25-(-11+19-4)]$

$37+[-25-(8-4)]$

$37+[-25-(4)]$

$37+[-29]$

8

E) $60\div\{2\cdot[-7+18\div(-3+12)]\}-[7\cdot(-3)-18\div(-2)+1]$

$60\div\{2\cdot[-7+18\div 9]\}-[-21-(-9)+1]$

$60\div\{2\cdot[-7+2]\}-[-12+1]$

$60\div\{2\cdot(-5)\}-(-11)$

$60\div(-10)-(-11)$

$-6+11$

5

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(2)

F) $2+8-3-5+15$

$10-3-5+15$

$7-5+15$

$2+15$

17

G) $\frac{2}{3}-\frac{1}{4}+\frac{1}{5}$

MMC (3,4,5)

3,4,5	2
3,2,5	2
3,1,5	3
1,1,5	5
1,1,1	60

$\frac{2}{3}=\frac{2\times 20}{60}=\frac{40}{60}$

$\frac{1}{4}=\frac{1\times 15}{60}=\frac{15}{60}$

$\frac{1}{5}=\frac{1\times 12}{60}=\frac{12}{60}$

$\frac{40}{60}-\frac{15}{60}+\frac{12}{60}=$

$\frac{(40-15+12)}{60}=\boxed{\frac{37}{60}}$

H) $(-3)\cdot(-2)$

6

i) $\frac{2}{3}\cdot\frac{1}{4}=\frac{2}{12}$

$\frac{2}{12}=\boxed{\frac{1}{6}}$

J) $\sqrt[3]{27}-\sqrt{81}-\sqrt{25}$

$3-9-5$

$-6-5$

-11

K) $3^4=$

$3\times3\times3\times3=81$

L) $\frac{1}{4}:\frac{1}{8}=$

$\frac{1}{4}\cdot\frac{8}{1}=\frac{1\cdot8}{4\cdot1}=\frac{8}{4}$

$\frac{8}{4}=2$

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M) $\left(\frac{1}{4}\right)^2$

$\left(\frac{1}{4}\right)^2 = \frac{1^2}{4^2} = \frac{1}{16}$ //

N) 2^{-4}

$2^{-4} = \frac{1}{2^4}$

$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$ // $\frac{1}{16}$

O) $\frac{2}{4} \cdot \frac{3}{5} + \frac{2}{4} : \frac{1}{3} - \left(\frac{2}{3}\right)^2$

$\frac{1}{2} \cdot \frac{3}{5} = \frac{1 \cdot 3}{2 \cdot 5} = \frac{3}{10}$ | $\frac{1}{2} : \frac{1}{3} = \frac{1}{2} \cdot \frac{3}{1} = \frac{3}{2}$ | $\frac{2^2}{3^2} = \frac{4}{9}$

$\frac{3}{10} + \frac{3}{2} - \frac{4}{9}$

MMC (10, 2, 9) | 2
5, 1, 9 | 3
5, 1, 3 | 3
5, 1, 1 | 5
1, 1, 1 | // 90

$\frac{3}{10} = 90 \div 10 = 9 \cdot 3 = \boxed{\frac{27}{90}}$

$\frac{3}{2} = 90 \div 2 = 45 \cdot 3 = \boxed{\frac{135}{90}}$

$\frac{4}{9} = 90 \div 9 = 10 \cdot 4 = \boxed{\frac{40}{90}}$

$\frac{27}{90} + \frac{135}{90} - \frac{40}{90} = \frac{27 + 135 - 40}{90} = \frac{162 - 40}{90} = \frac{122 \div 2}{90 \div 2} = \frac{61}{45}$ //

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(3)

A) $3^4 \cdot 3^5 \div 3^2$

$3^{4+5} = 3^9 \div 3^2 = 3^{9-2} = 3^7$ //

B) $\dfrac{2^{15} \cdot 2^4 \cdot 2^6}{2^5 \cdot 2^7} = \dfrac{2^{25}}{2^{12}} = 2^{25-12} = 2^{13}$ //

C) $(3^3)^4 \cdot 3^4 \div (3^2 \cdot 3^5)$

$3^{12} \cdot 3^4 \qquad 3^{2+5}$

$3^{16} \div \qquad 3^7$

$\dfrac{3^{16}}{3^7} = 3^{16-7} = 3^9$ //

(4)

A) $a^2 + b \cdot c$ $\qquad a = -1,\ b = 2$ e $c = -3$

$(-1)^2 + 2 \cdot (-3)$

$1 + -6$

-5 //

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4

B) $\sqrt{b^2 - 4 . a . c}$ $a = 1,\ b = -4,\ c = 3$

$\sqrt{(-4)^2 - 4 . (1) . (3)}$

$\sqrt{16 - 12} = \sqrt{4}$

$\sqrt{4} = 2$ //

c) $\dfrac{x - 3 y x z}{x + y . z}$ $x = -2,\ y = 3,\ z = -1$

$\dfrac{(-2) - 3 . 3 . (-2) . (-1)}{(-2) + (3) . (-1)} = \dfrac{(-2) - 18 = -20}{-2 + (-3) = -5}$

$\dfrac{-20}{-5} = 4$ //

5

A) $8x - 11 = 12 - 2x - 2$

$12 - 2x - 2 = 10 - 2x$

$8x - 11 = 10 - 2x$

$8x + 2x - 11 = 10$

$10x - 11 = 10$

$x = \dfrac{21}{10}$ //

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(5)

B) $5x - (8+10) = 2.(6x-4) - 2$

$5x - 18 = 2.(6x-4) - 2$

$5x - 18 = 12x - 8 - 2$

$5x - 18 = 12x - 10$

$-18 = 7x - 10$

$-8 = 7x$

$x = -\frac{8}{7}$ //

c) $\frac{x-2}{3} + \frac{2x+3}{2} = 12 - 5$

$\frac{x-2}{3} + \frac{2x+3}{2} = 7$

MMC (3, 2) | 2
3, 1 | 3
1, 1 | 6

$6.\left(\frac{x-2}{3}\right) + 6.\left(\frac{2x+3}{2}\right) = 6.7$

$2(x-2) + 3(2x+3) = 42$

$2x - 4 + 6x + 9 = 42$

$8x + 5 = 42$

$8x = 37$

$x = \frac{37}{8}$ //

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(5)

D) $\frac{x+3}{6} + \frac{x-3}{2} = \frac{x}{2} + \frac{x+2}{3}$

MMC (6, 2, 3) | 2
3, 1, 3 | 3
1, 1, 1 | 6

$6 \cdot \left(\frac{x+3}{6}\right) + 6 \cdot \left(\frac{x-3}{2}\right) = 6 \cdot \left(\frac{x}{2}\right) + 6 \cdot \left(\frac{x+2}{3}\right)$

$(x+3) + 3(x-3) = 3x + 2(x+2)$

$x + 3 + 3x - 9 = 4x - 6$

$3x + 2x + 4 = 5x + 4$

$4x - 6 = 5x + 4$

$-6 = x + 4$

$x = -10$ //

(6)

A) $F(x): y = 3 - x$

X	Y
1	2
2	1
3	0
4	-1
-1	4

B) $F(x): y = x + 2$

X	Y
-1	1
-3	-1
-2	0
-4	-2
-5	-3

C) $F(x) = y = x - 1$

X	Y
0	-1
0	-1
0	-1
0	-1
0	-1

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7 - DETERMINE OS ZEROS E O VÉRTICE DAS FUNÇÕES:

A) $y = x^2 - 5x + 6$

ZEROS DA FUNÇÃO = 3 e 2

$\Delta = b^2 - 4ac \rightarrow \Delta = (-5)^2 - 4(1)(6) = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} \rightarrow x = \frac{-(-5) \pm \sqrt{1}}{2(1)} = \frac{5 \pm 1}{2}$

$x_1 = \frac{5+1}{2} = \boxed{3}$

$x_2 = \frac{5-1}{2} = \boxed{2}$

VÉRTICES DA FUNÇÃO = 2,5 e -0,25

$x_v = \frac{-b}{2a} \rightarrow x_v = \frac{-(-5)}{2(1)} = \boxed{2,5}$

$y_v = \frac{-\Delta}{4a} \rightarrow y_v = \frac{-1}{4(1)} = \boxed{-0,25}$

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(7) B) $y = 4x^2 - 16$

ZEROS DA FUNÇÃO = $\boxed{-2 \text{ e } 2}$

$4x^2 - 16 = 0$

$4x^2 = 16$

$x^2 = \frac{16}{4}$

$x^2 = 4$;

$x = \pm\sqrt{4}$;

VÉRTICES DA FUNÇÃO = $\boxed{0 \text{ e } -16}$

$a = 4$, $b = 0$ e $c = -16$

$x_v = \frac{-b}{2a} \rightarrow x_v = \frac{-0}{2(4)} = 0$

$y_v = -16$

C) $y = x^2 + 4$

ZEROS DA FUNÇÃO = [redacted]

$x^2 + 4 = 0$

$x^2 = -4$

$x = \pm\sqrt{-4}$

VÉRTICES DA FUNÇÃO = $\boxed{0 \text{ e } 4}$

$a = 1$, $b = 0$ e $c = 4$

$x_v = \frac{-b}{2a} \rightarrow x_v = \frac{-0}{2(1)} = 0$

$y_v = 4$

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(8) - DETERMINE A ALTURA DE H

$$3 = a(4)^2 + H \rightarrow 3 = 16a + H$$

$$3 = 16\left(\frac{-H}{25}\right) + H$$

$$3 = \frac{-16H}{25} + H$$

$$3 = \frac{-16H}{25} + \frac{25H}{25}$$

$$3 = \frac{9H}{25}$$

$$75 = 9H$$

$$H = \frac{75}{9}$$

$$H = \frac{25}{3} \simeq \boxed{8{,}33 \text{ METROS DE ALTURA}}$$

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